

CSA3006 - DATA MINING AND DATA WAREHOUSING

Presented By

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Module V

- **Advanced Data Mining-** Regression and time series processing. Nonlinear regression models. Regression trees. Radial Basis Networks. Time series analysis (trend analysis, pattern detection, prediction models, anomaly detection). Ensemble methods. Voting. Bagging. Boosting (AdaBoost). Random forests. Stacking. Mining sequence patterns in Transactional databases– Mining Object– Spatial– Multimedia–Text and Web data – Spatial Data mining– Multimedia Data mining–Text Mining– Mining the World Wide Web. Document classification and clustering.
- **Case Study:** Mining Real data- Pre-processing data from a real medical domain. Applying various data mining techniques to create a comprehensive and accurate model of the data.

Radial Basis Networks

- Radial Basis Function (RBF) Neural Networks are used for function approximation tasks.
- They are a special category of feed-forward neural networks comprising of three layers. Due to this distinct three-layer architecture and universal approximation capabilities they offer faster learning speeds and efficient performance in classification and regression problems.

How Do RBF Networks Work?

- RBF Networks are conceptually similar to K-Nearest Neighbor (k-NN) models though their implementation is distinct. The fundamental idea is that nearby items with similar predictor variable values influence an item's predicted target value. Here's how RBF Networks operate:
- **Input Vector:** The network receives an n-dimensional input vector that needs classification or regression.
- **RBF Neurons:** Each neuron in the hidden layer represents a prototype vector from the training set. The network computes the Euclidean distance between the input vector and each neuron's center.
- **Activation Function:** The Euclidean distance is transformed using a Radial Basis Function (typically a Gaussian function) to compute the neuron's activation value. This value decreases exponentially as the distance increases.
- **Output Nodes:** Each output node calculates a score based on a weighted sum of the activation values from all RBF neurons. For classification the category with the highest score is chosen.

Key Characteristics of RBFs

- **Radial Basis Functions:** These are real-valued functions dependent solely on the distance from a central point. The Gaussian function is the most commonly used type.
- **Dimensionality:** The network's dimensions correspond to the number of predictor variables.
- **Center and Radius:** Each RBF neuron has a center and a radius (spread). The radius affects how broadly each neuron influences the input space.

Architecture of RBF Networks

- The architecture of an RBF Network typically consists of three layers:

1. Input Layer

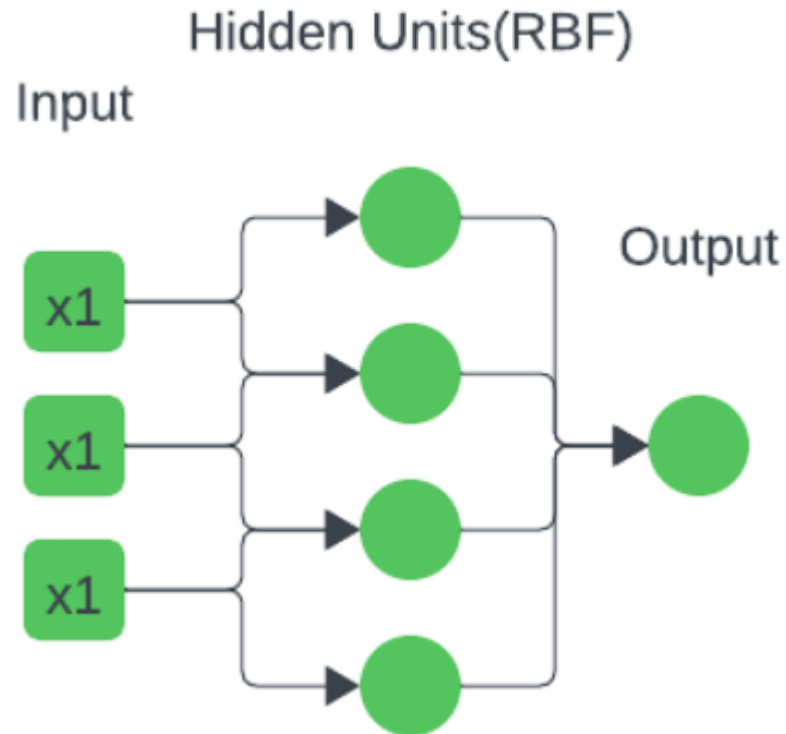
- **Function:** After receiving the input features the input layer sends them straight to the hidden layer.
- **Components:** It is made up of the same number of neurons as the characteristics in the input data. One feature of the input vector corresponds to each neuron in the input layer.

2. Hidden Layer

- **Function:** This layer uses radial basis functions (RBFs) to conduct the non-linear transformation of the input data.
- **Components:** Neurons in the buried layer apply the RBF to the incoming data. The Gaussian function is the RBF that is most frequently utilized.
- **RBF Neurons:** Every neuron in the hidden layer has a spread parameter (σ) and a center which are also referred to as prototype vectors. The spread parameter modulates the distance between the center of an RBF neuron and the input vector which in turn determines the neuron's output.

3. Output Layer

- **Function:** The output layer uses weighted sums to integrate the hidden layer neurons outputs to create the network's final output.
- **Components:** It is made up of neurons that combine the outputs of the hidden layer in a linear fashion. To reduce the error between the network's predictions and the actual target values, the weights of these combinations are changed during training.



Regression Trees

- A **Regression Tree** is a type of decision tree used for predicting **continuous numerical values** (e.g., price, temperature, salary).
- It splits the dataset into smaller regions and predicts the output as the **average (mean)** of observations in each region.
- Instead of classification labels, regression trees predict **real-valued outputs**.
- The dataset is recursively divided based on feature values.
- Each final node (**leaf node**) contains a predicted numeric value.

Working of Regression Trees

1. Start with Entire Dataset

- Consider all data points as one region.

2. Select Best Split

- Choose a feature and split point that minimizes error.
- Typically uses **Mean Squared Error (MSE)**.

3. Recursive Splitting

- Repeat splitting for each subset until stopping criteria are met.

4. Assign Prediction

- Each leaf node outputs the **mean value** of target variable in that region.

Splitting Criterion

- Regression trees aim to minimize variance using:

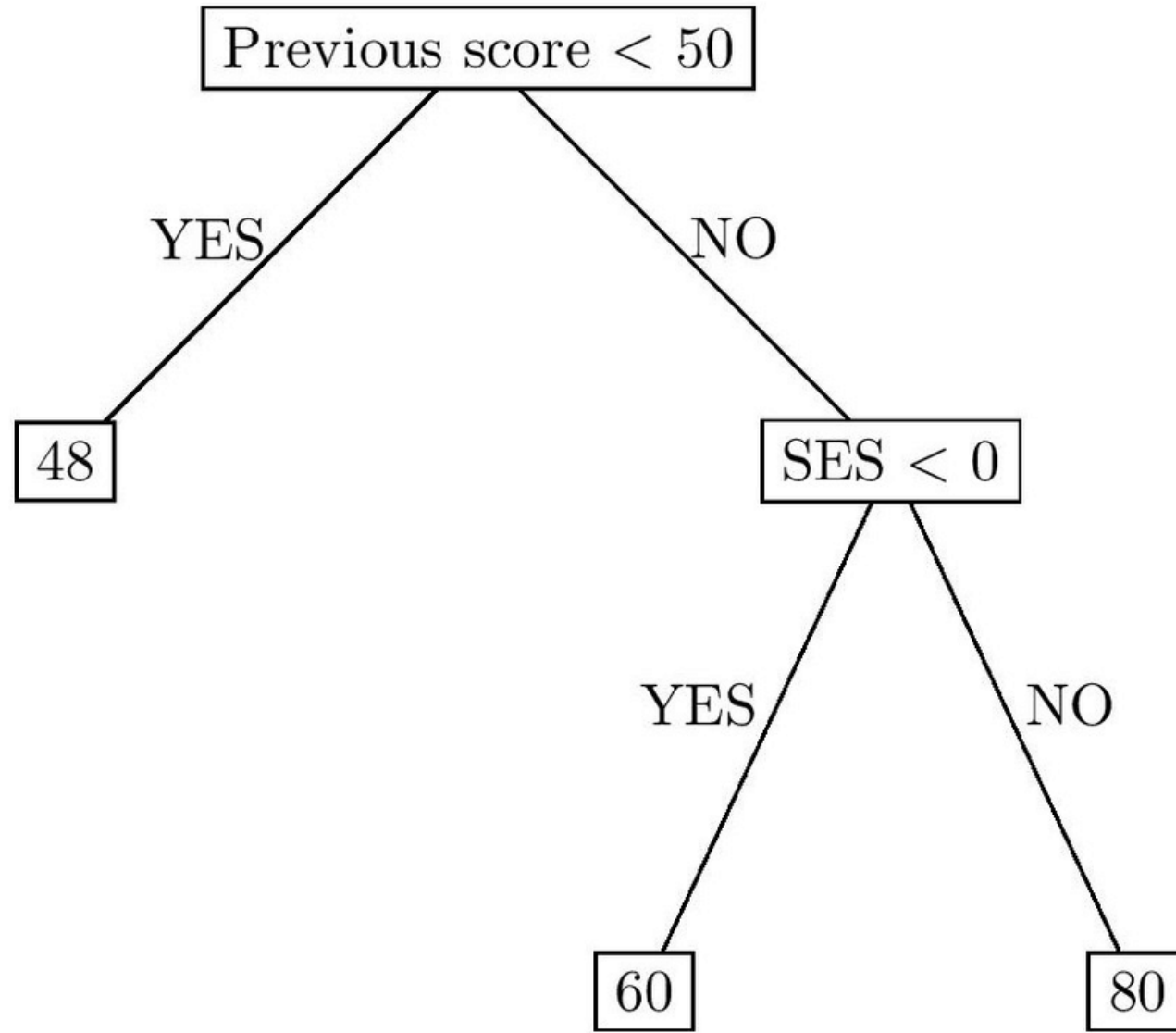
$$MSE = \frac{1}{n} \sum (y_i - \bar{y})^2$$

Where:

- y_i = actual value
 - $\bar{y}$ = mean of values in the node
- The best split is the one that gives **lowest total MSE**.

Decision Tree Regressor Components

- Root Node: The initial, top-level node that splits the entire dataset.
- Leaf Node: The final nodes that contain the prediction (numerical output).
- Branches: Connect nodes based on decision rules (e.g., Temperature < 30).



Pruning in Regression Trees

To avoid overfitting:

- **Pre-pruning:** Stop tree growth early
- **Post-pruning:** Remove unnecessary branches

Advantages

- Simple and easy to understand
- Handles non-linear relationships
- No need for data normalization
- Works with both numerical & categorical data

Disadvantages

- Prone to overfitting
- Sensitive to small data changes
- Produces step-wise predictions (not smooth)

Applications

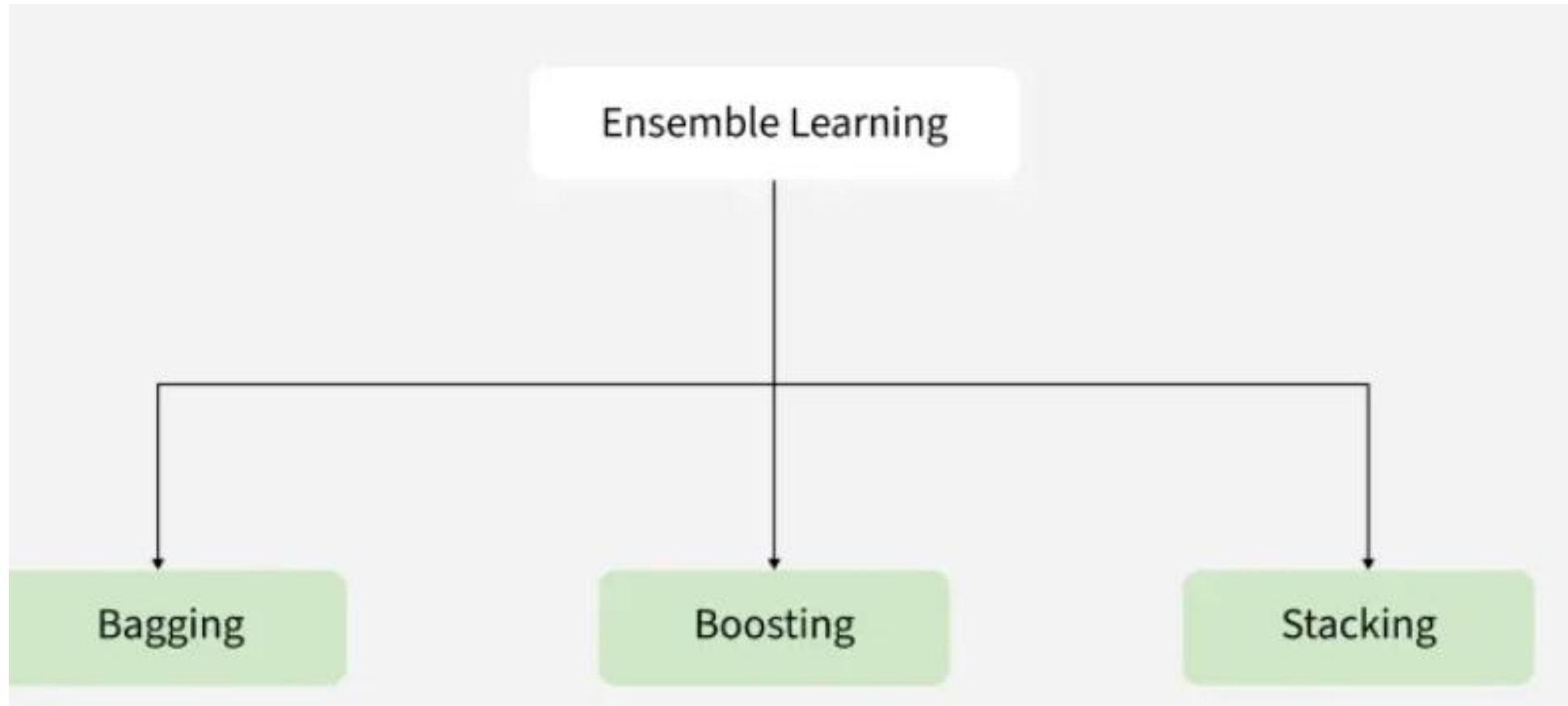
- House price prediction
- Stock value estimation
- Sales forecasting
- Risk assessment

Ensemble Learning Model

- Ensemble learning is a method *where we use many small models instead of just one.*
- Each of these models may not be very strong on its own, but when we put their results together, *we get a better and more accurate answer.*
- *It's like asking a group of people for advice instead of just one person—each one might be a little wrong, but together, they usually give a better answer.*

- Ensemble learning is a method where multiple models are combined instead of using just one. Even if individual models are weak, combining their results gives more accurate and reliable predictions.
- **Multiple Models:** Uses many small models together
- **Better Accuracy:** Combined results improve performance
- **Reduced Errors:** Mistakes of one model are balanced by those of others
- **Simple Idea:** Like taking advice from a group instead of one person

Types of Ensembles Learning

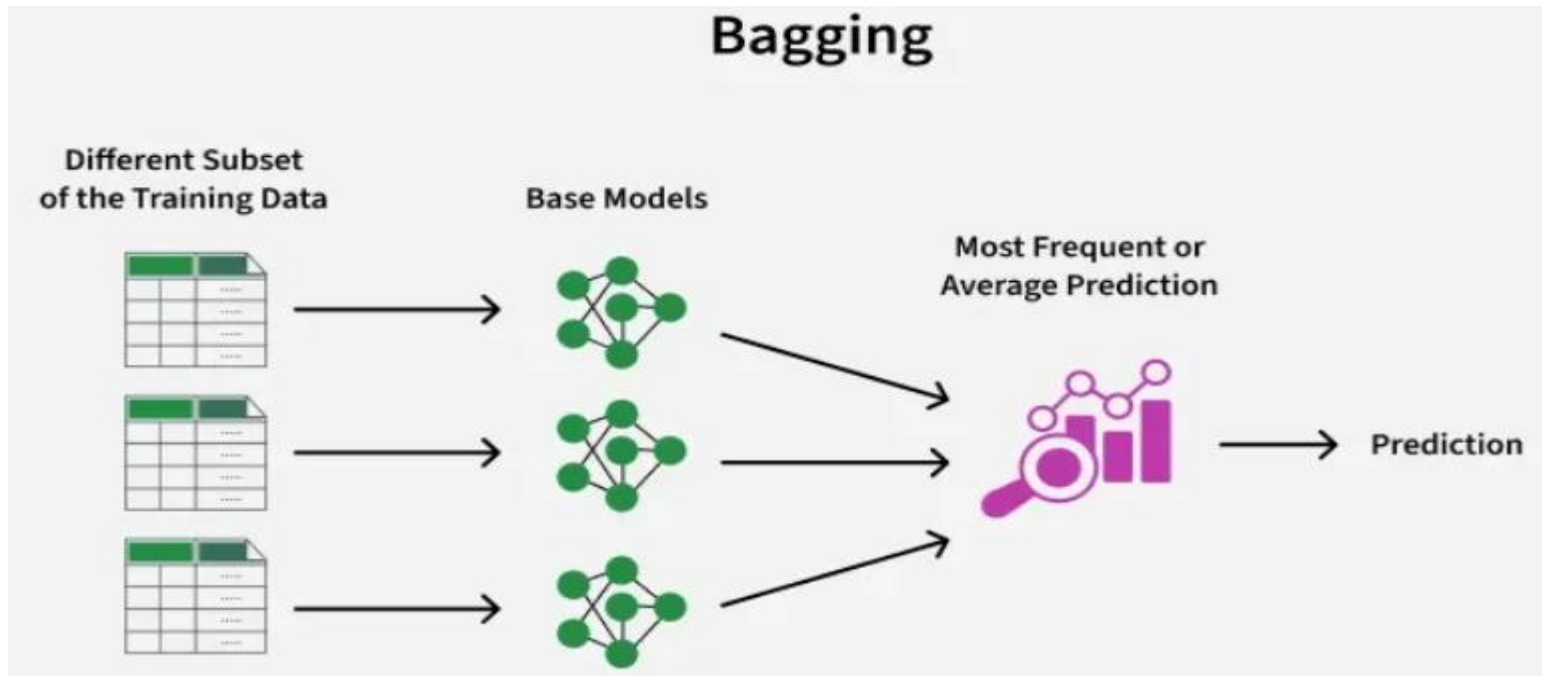


Benefits of Ensemble Learning

- Ensemble learning is a versatile approach that can be applied to machine learning model for:
- **Reduction in Overfitting:** By aggregating predictions of multiple model's ensembles can reduce overfitting that individual complex models might exhibit.
- **Improved Generalization:** It generalizes better to unseen data by minimizing variance and bias.
- **Increased Accuracy:** Combining multiple models gives higher predictive accuracy.
- **Robustness to Noise:** It mitigates the effect of noisy or incorrect data points by averaging out predictions from diverse models.
- **Flexibility:** It can work with diverse models including decision trees, neural networks and support vector machines making them highly adaptable.
- **Bias-Variance Tradeoff:** Techniques like bagging reduce variance, while boosting reduces bias leading to better overall performance.

1. Bagging (Bootstrap Aggregating):

- Models are trained independently on *different random subsets of the training data*.
- Their results are then combined, usually by *averaging (for regression) or voting (for classification)*.
- This helps *reduce variance and prevents overfitting*.



Bagging Algorithm

- It can be used for *both regression and classification tasks*.

1. Bootstrap Sampling:

- Divides the original training data into 'N' subsets and randomly selects a subset with replacement in some rows from other subsets.
- This step ensures that the base models are trained on diverse subsets of the data and there is no class imbalance.

2. Base Model Training:

- For each bootstrapped sample we train a base model independently on that subset of data.
- These weak models are trained in parallel to increase computational efficiency and reduce time consumption.
- We can use different base learners i.e. different ML models as base learners to bring variety and robustness.

3. Prediction Aggregation:

- To make a prediction on testing data combine the predictions of all base models.
- For *classification tasks it can include majority voting* or weighted majority while for *regression it involves averaging the predictions*.

4. Out-of-Bag (OOB) Evaluation:

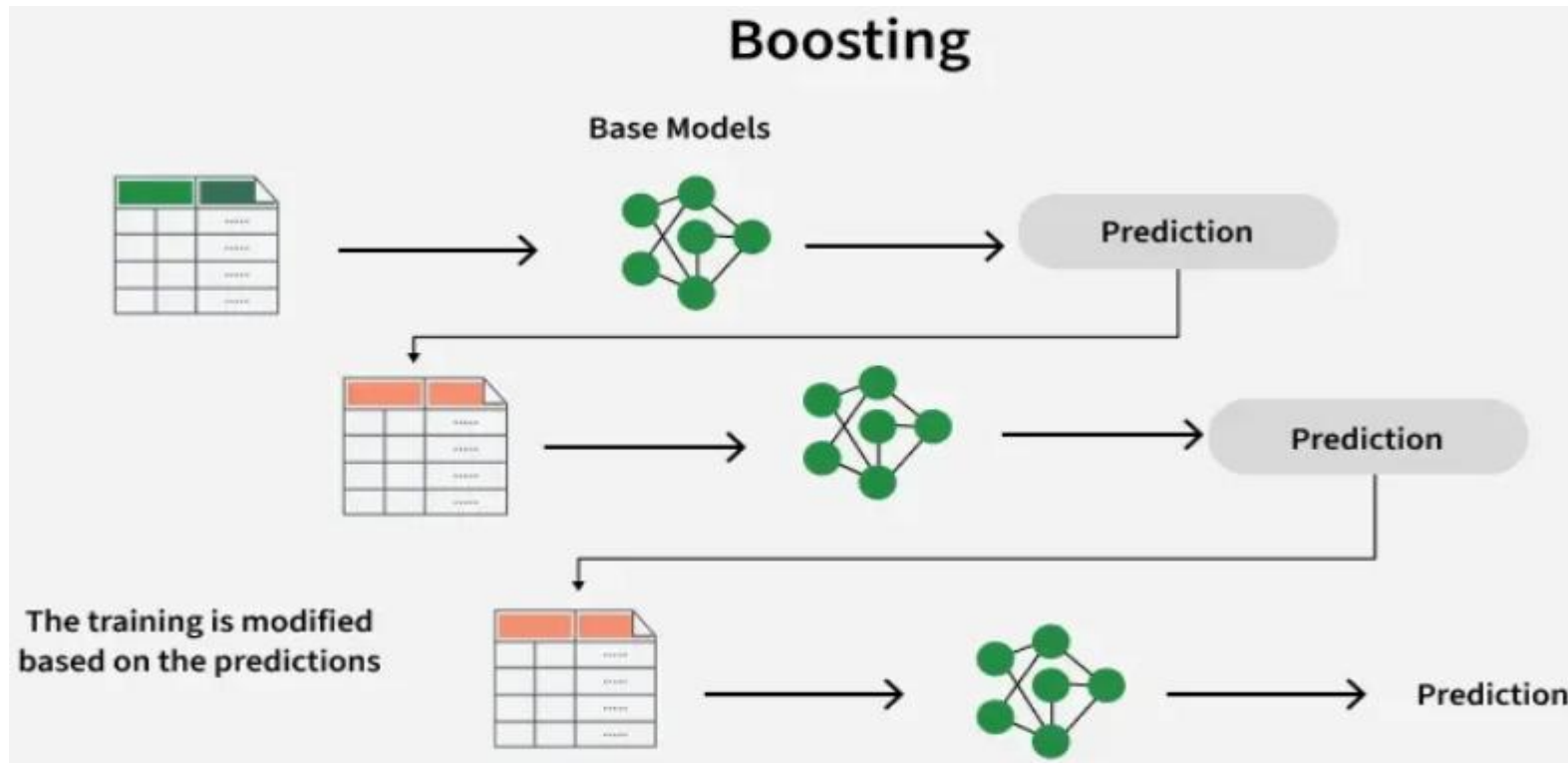
- Some samples are excluded from the training subset of particular base models during the bootstrapping method.
- These “out-of-bag” samples can be used to estimate the model’s performance without the need for cross-validation.

5. Final Prediction:

After aggregating the predictions from all the base models, Bagging produces a final prediction for each instance.

2. Boosting:

- Models are trained one after another.
- Each new model focuses on fixing the errors made by the previous ones.
- The final prediction is a weighted combination of all models, which helps reduce bias and improve accuracy.

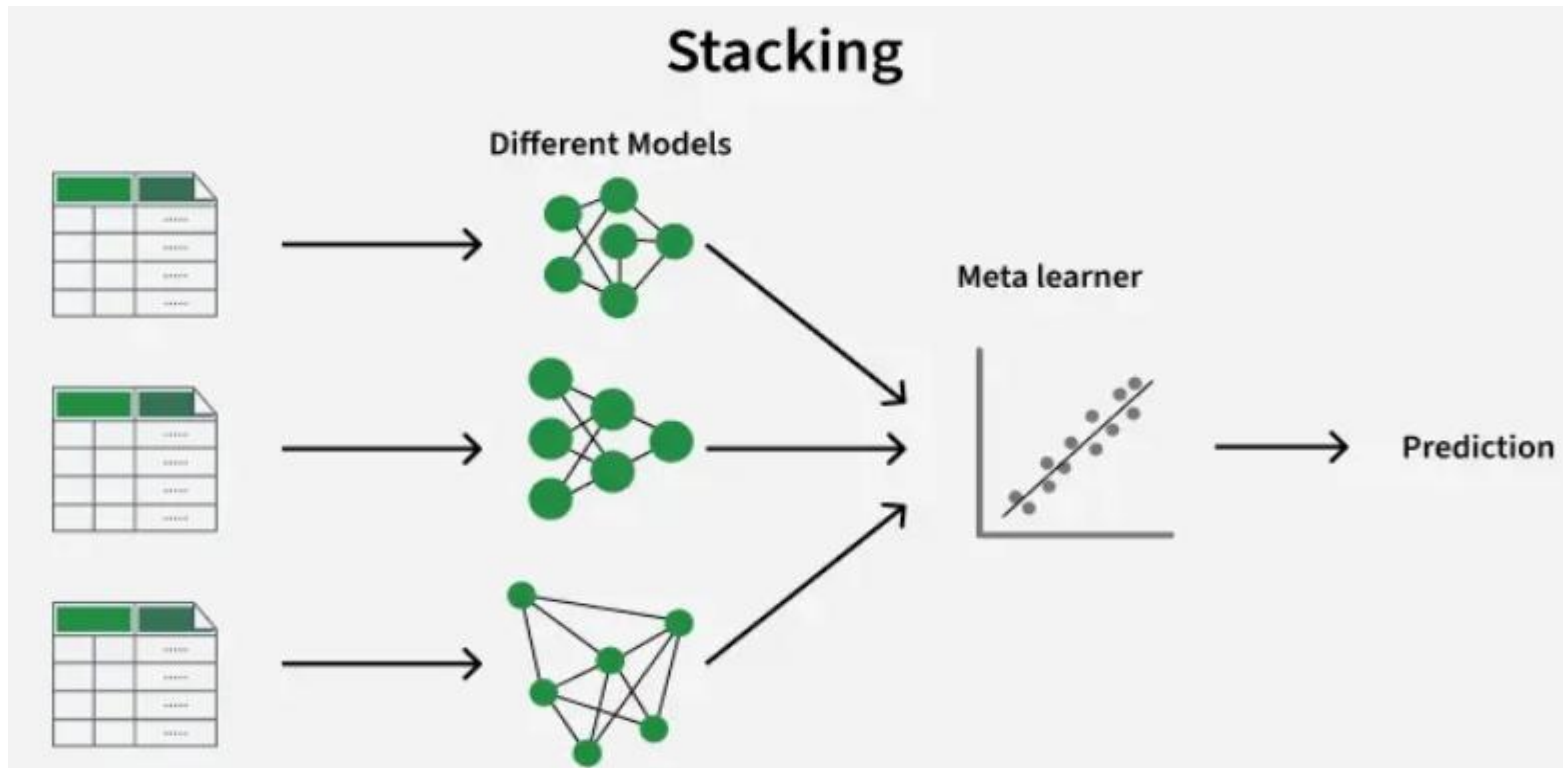


Boosting Algorithm

- Boosting is an ensemble technique where multiple weak models are trained one after another, and each new model focuses on correcting the errors of the previous one to build a strong model. The process works as follows:
- **Initialize Weights** : Start with equal weights for all training data
- **Train Weak Learner** : Train a simple model on the dataset
- **Sequential Learning** : Each new model learns from previous errors
- **Weight Adjustment** : Misclassified samples get higher weights so future models focus more on them

3. Stacking (Stacked Generalization):

- Multiple different models (often of different types) are trained, and their *predictions are used as inputs to a final model, called a meta-model*.
- The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.

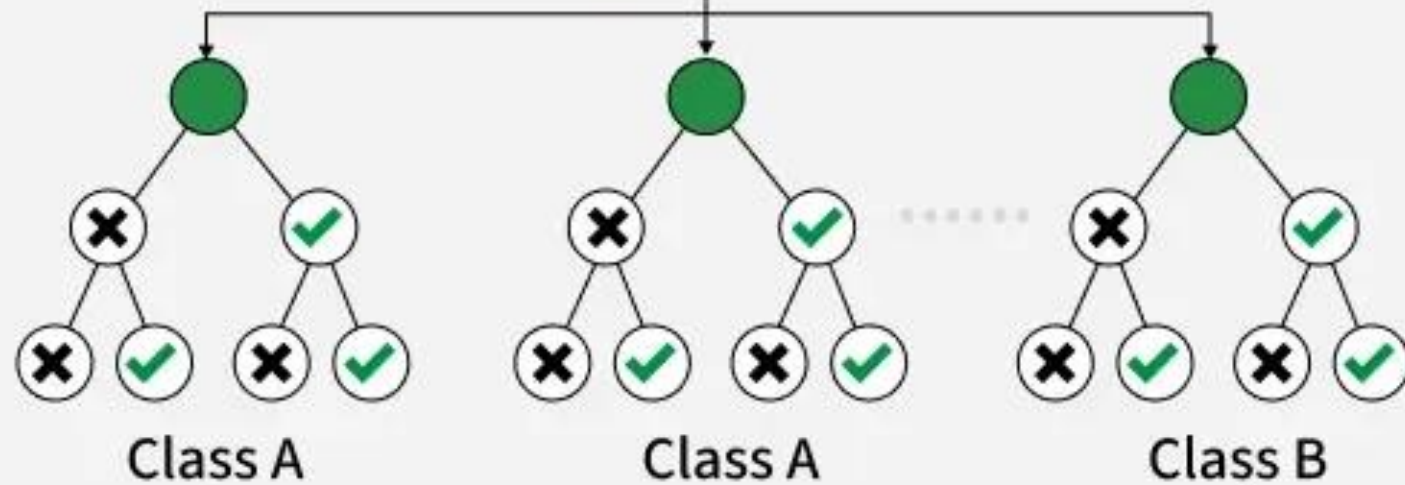


Random Forests

- Random Forest is a machine learning algorithm that uses many decision trees to make better predictions.
- Each tree looks at different random parts of the data and their results are combined by voting for classification or averaging for regression which makes it as ensemble learning technique.
- This helps in improving accuracy and reducing errors.

Training Data Instance

Model
Training



Decision
Trees

Model
Testing

Bagging (Voting Majority)

Prediction Output

Class A

Random Forest Effectiveness

High Accuracy

Provides reliable and precise predictions



Handles Missing Values

The algorithm's robustness to incomplete data



Works with Large Datasets

Efficient processing of high-dimensional data



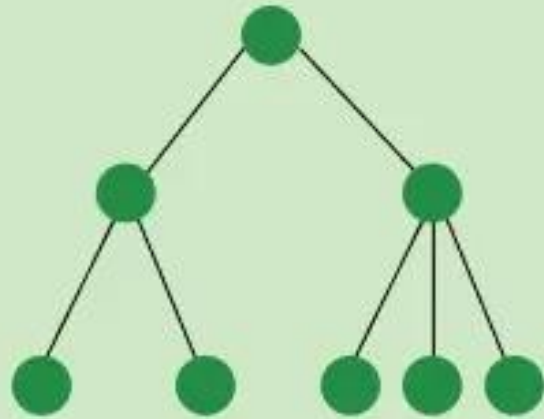
Reduces Overfitting

Ensures generalization to unseen data



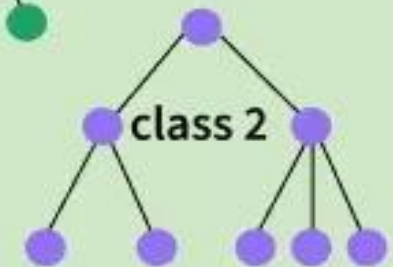
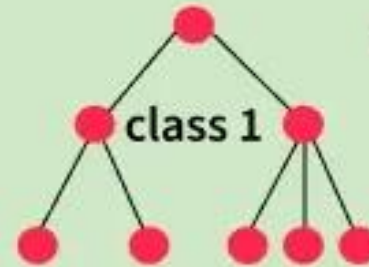
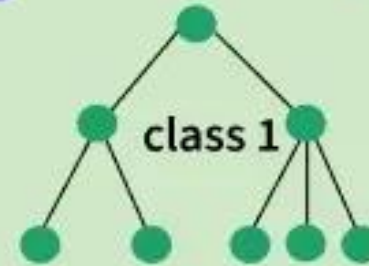
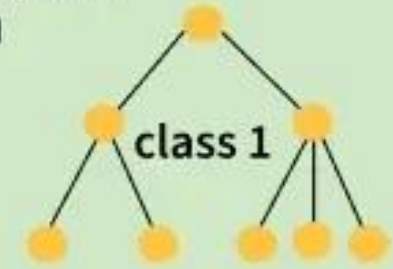
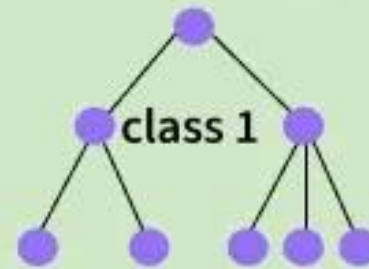
Single Decision Tree

Ensemble of trees for more accurate and robust prediction



Random Forest

Prediction from a single decision path



Working of Random Forest Algorithm

- **Create Many Decision Trees:** The algorithm makes many decision trees each using a random part of the data. So every tree is a bit different.
- **Pick Random Features:** When building each tree it doesn't look at all the features (columns) at once. It picks a few at random to decide how to split the data. This helps the trees stay different from each other.
- **Each Tree Makes a Prediction:** Every tree gives its own answer or prediction based on what it learned from its part of the data.
- **Combine the Predictions:** For classification, the final answer is the category that most trees vote for (majority voting).
- **Why It Works Well:** Using random data and features for each tree helps avoid overfitting and makes the overall prediction more accurate and trustworthy.

Key Features of Random Forest

- **Handles Missing Data:** It can work even if some data is missing so you don't always need to fill in the gaps yourself.
- **Shows Feature Importance:** It tells you which features (columns) are most useful for making predictions which helps you understand your data better.
- **Works Well with Big and Complex Data:** It can handle large datasets with many features without slowing down or losing accuracy.
- **Used for Different Tasks:** You can use it for both **classification** like predicting types or labels and **regression** like predicting numbers or amounts.

Assumptions of Random Forest

- **Each tree makes its own decisions:** Every tree in the forest makes its own predictions without relying on others.
- **Random parts of the data are used:** Each tree is built using random samples and features to reduce mistakes.
- **Enough data is needed:** Sufficient data ensures the trees are different and learn unique patterns and variety.
- **Different predictions improve accuracy:** Combining the predictions from different trees leads to a more accurate final result.

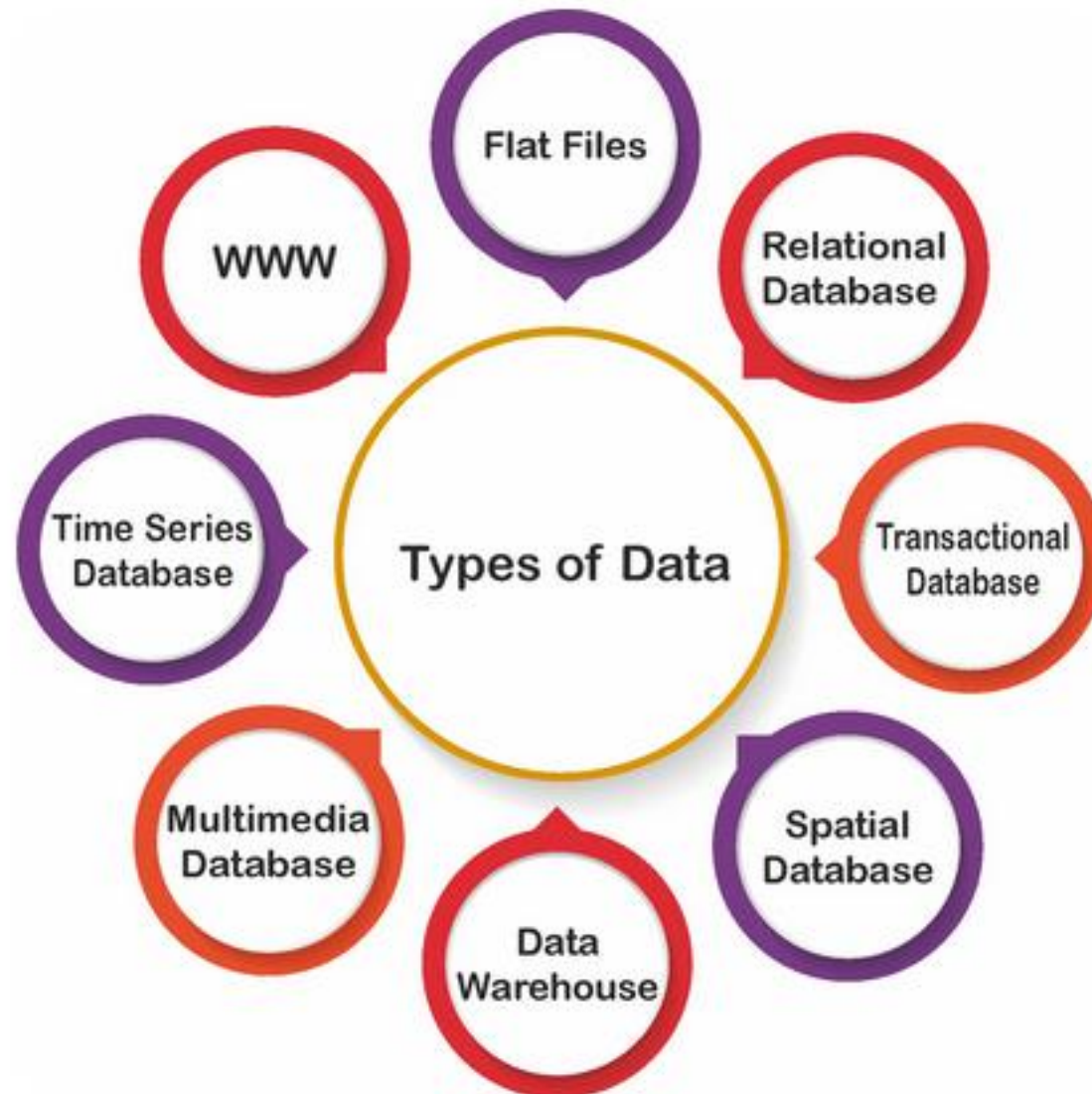
Advantages of Random Forest

- Random Forest provides very accurate predictions even with large datasets.
- Random Forest can handle missing data well without compromising with accuracy.
- It doesn't require normalization or standardization on dataset.
- When we combine multiple decision trees it reduces the risk of overfitting of the model.

Limitations of Random Forest

- It can be computationally expensive especially with a large number of trees.
- It's harder to interpret the model compared to simpler models like decision trees.

Data mining on different kinds of data



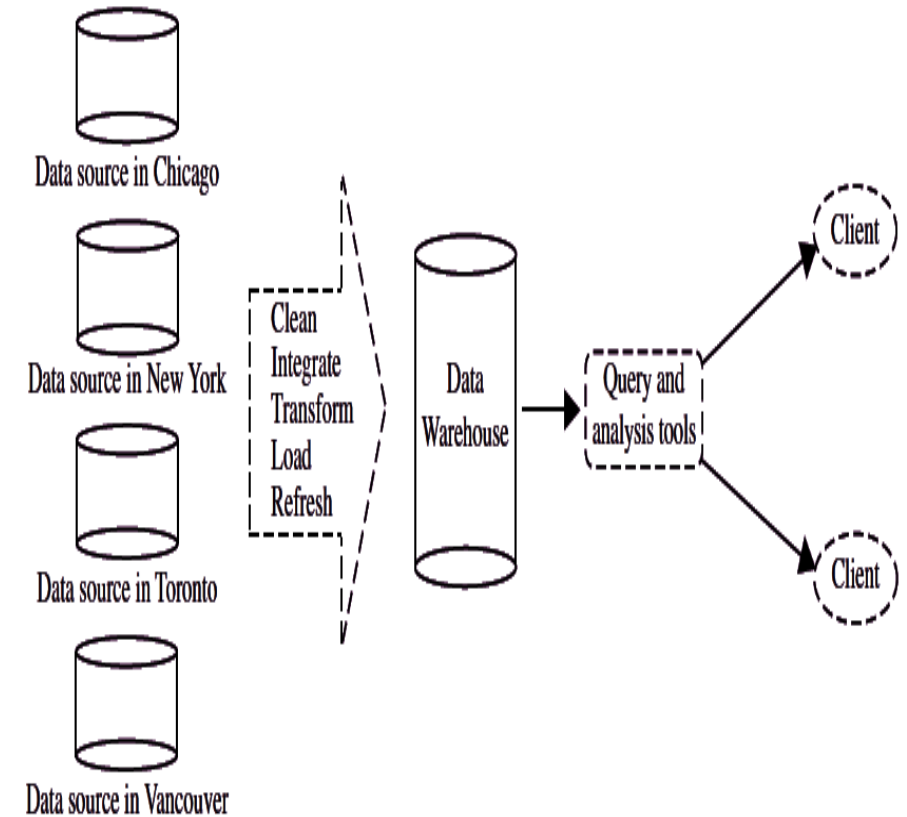
- Data mining techniques are applied to **various types of data**, depending on the data source, structure, and application domain. Each kind of data requires **specialized mining methods**.

- **1. Relational Database Data**
- Data stored in **tables (rows and columns)**.
- Common in traditional database systems.
- Mining techniques:
 - Classification
 - Association rule mining
 - Clustering
- Example: Student records, sales databases.

The diagram illustrates a table with four columns: Name, Age, Gender, and City. The rows represent individual records. Annotations include: a blue arrow labeled 'Column' pointing to the 'City' header; a red arrow labeled 'Field' pointing to the '25' value in the 'Age' column for the first row; and a green arrow labeled 'Row' pointing to the entire first row (Akhil, 25, Male, Hyderabad).

Name	Age	Gender	City
Akhil	25	Male	Hyderabad
Sai	25	Male	Mumbai
Varsha	28	Female	Chennai
Bindu	20	Female	Delhi

- **2. Data Warehouse Data**
- Integrated, subject-oriented, historical data.
- Supports **multidimensional analysis (OLAP)**.
- Mining focuses on:
 - Trends
 - Patterns over time
 - Aggregations
- Example: Business intelligence and decision support systems.



- **3. Transactional Data**
- Consists of **transactions** with item sets.
- Widely used in **market basket analysis**.
- Mining tasks:
 - Frequent pattern mining
 - Association rule discovery
- Example: Supermarket purchase transactions.

TID Items

T1 Bread, Coke, Milk

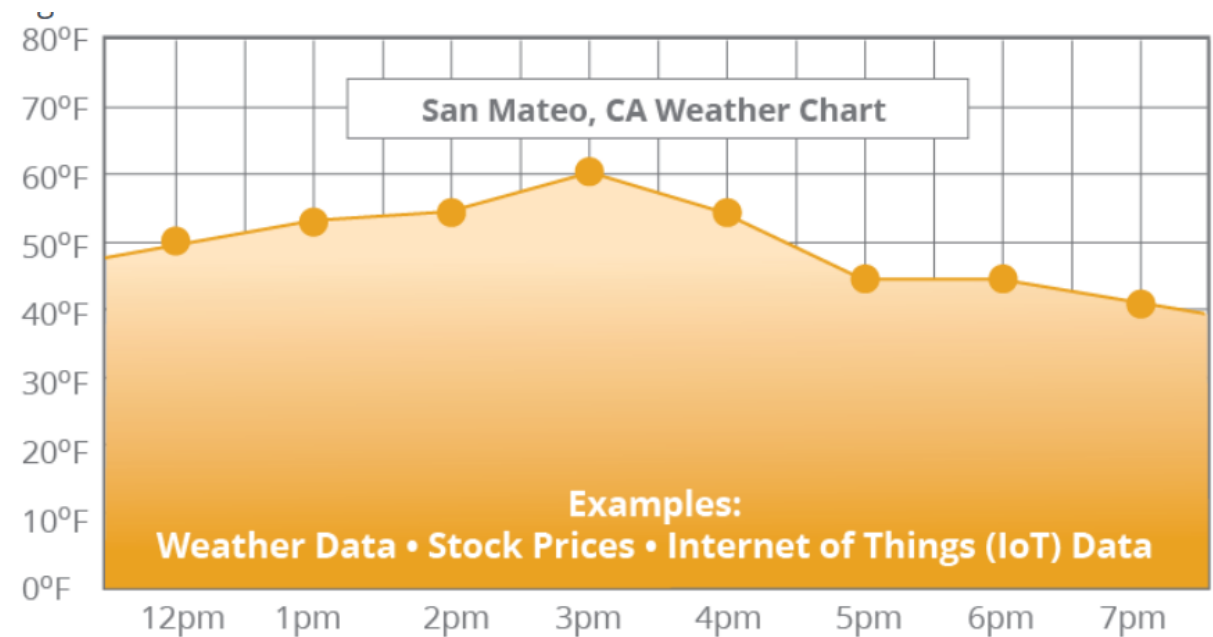
T2 Popcorn, Bread

T3 Popcorn, Coke, Egg, Milk

T4 Popcorn, Bread, Egg, Milk

T5 Coke, Egg, Milk

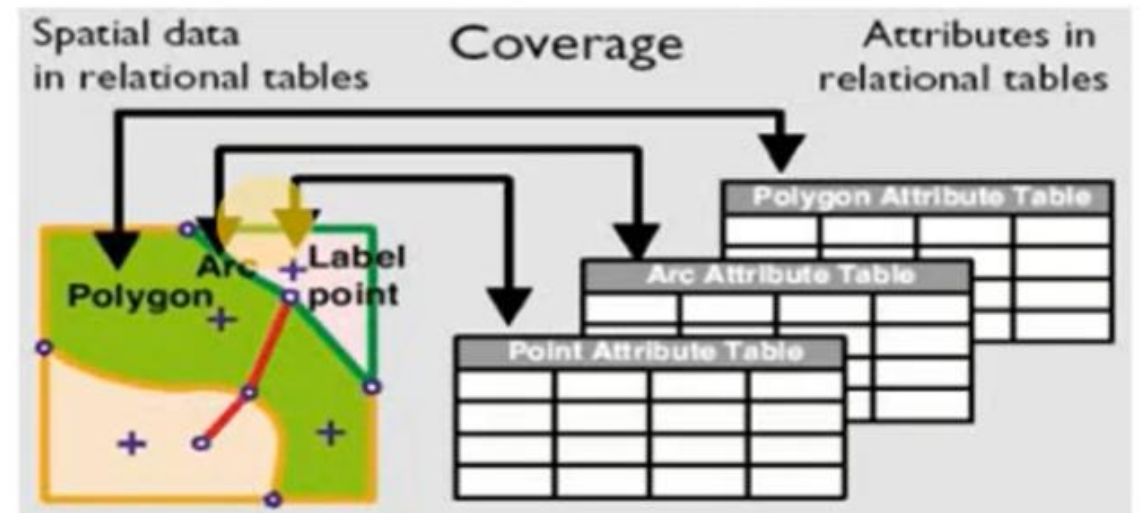
- **4. Time-Series Data**
- Data recorded over **time intervals**.
- Important for forecasting and trend analysis.
- Mining techniques:
 - Trend analysis
 - Similarity search
 - Sequential pattern mining
- Example: Stock prices, weather data, ECG signals.



- **5. Sequence Data**

- Ordered list of events or symbols.
- Focuses on discovering **frequent sequences**.
- Mining tasks:
 - Sequential pattern mining
 - Episode discovery
- Example: Web click streams, DNA sequences.

- **6. Spatial Data**
- Data related to **geographical locations**.
- Includes spatial relationships like distance and direction.
- Mining techniques:
 - Spatial clustering
 - Spatial association rules
- Example: GIS data, satellite images.



- **7. Spatio-Temporal Data**
- Combines **spatial and time-based** information.
- Mining tasks:
 - Movement pattern discovery
 - Trajectory analysis
- Example: GPS tracking, traffic monitoring systems.

- **8. Text Data**

- Unstructured or semi-structured textual data.
- Requires text preprocessing.
- Mining techniques:
 - Text classification
 - Topic modeling
 - Sentiment analysis
- Example: Emails, research papers, social media posts.



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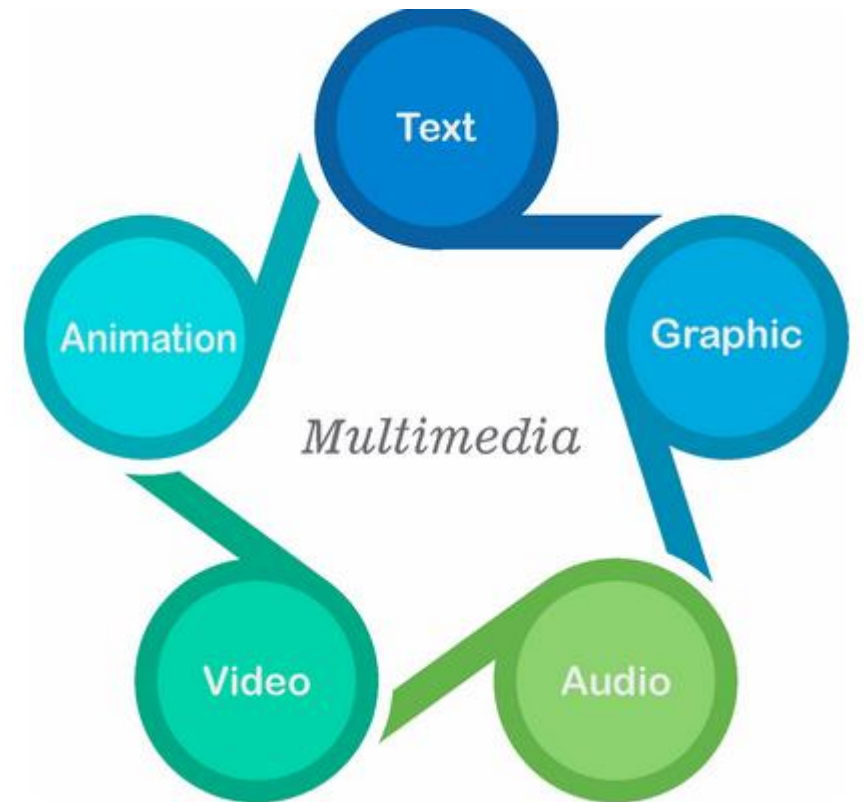
```
"OrderID", "CustomerID", "OrderDate"  
"01", "001", "06/06/2021"  
"02", "369", "06/06/2021"  
"03", "151", "06/06/2021"  
"04", "014", "06/06/2021"  
"05", "061", "06/06/2021"  
"06", "220", "06/06/2021"
```

- **9. Web Data**
- Data extracted from the World Wide Web.
- Types:
 - Web content data
 - Web structure data
 - Web usage data
- Mining tasks:
 - Web usage mining
 - Page ranking
 - Recommendation systems
- Example: E-commerce websites.



- **10. Multimedia Data**

- Includes images, audio, and video.
- Mining involves feature extraction.
- Mining techniques:
 - Image classification
 - Video pattern detection
 - Audio mining
- Example: Medical images, surveillance videos.



- **11. Graph and Network Data**
- Data represented as nodes and edges.
- Mining focuses on **relationships**.
- Mining techniques:
 - Community detection
 - Link analysis
- Example: Social networks, citation networks.

STRUCTURED VS UNSTRUCTURED DATA IN A NUTSHELL

Properties	Structured data	Unstructured data
Format examples	• CSV • Excel	• audio files (WAV, MP3, OGG) • PDF documents • images (JPEG, PNG, etc.)
Sources examples	• online forms • point-of-sale (POS) systems • online transaction processing (OLTP) systems	• emails • social media posts • multimedia files • IoT outputs
Nature of data	Quantitative	Qualitative
Databases	Relational (SQL)	Non-relational (NoSQL)
Storage for analytics use	Warehouses and data lakehouses	Data lakes and data lakehouses
Specialists to handle data	Business analysts, software engineers, data analysts	Data scientists, data engineers, data analysts
Main benefits	Easy to search and analyze, doesn't require much space	Easy to collect and store
Main challenges	All data must fit predefined schema	Difficult to search and analyze